

GLOBAL-LOCAL SALIENCY DECODING FOR MICRO-EXPRESSION SPOTTING IN LONG VIDEOS

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ABSTRACT

Long-video micro-expression spotting requires continuous temporal responses to be decoded into discrete events, yet peak amplitude alone does not describe either a candidate’s prominence relative to the response background or its local structure across scales. We propose the Global-Local Saliency Decoder (GLSD), a training-free candidate-level decoder with no learned parameters for frozen temporal responses. GLSD combines normalized global prominence with aligned local multi-scale saliency while retaining each pipeline’s native event-output rules. On official temporal responses from ME-TST+ and BoostingVRME, GLSD yields higher pooled Spotting F1 point estimates in all four pipeline-dataset settings on SAMMLV and CAS(ME)³, with relative gains of 5.58% – 18.15% over Native decoding. BoostingVRME with GLSD reaches 0.3088 and 0.1152, respectively. The locked-parameter fusion ablation shows that the two structural cues contribute jointly to the GLSD decoder. Our code is public at <https://kx111666.github.io>.

Index Terms— micro-expression spotting, temporal event decoding, peak prominence, multi-scale temporal structure, candidate scoring

1. INTRODUCTION

Micro-expressions (MEs) are brief and involuntary facial movements that can reveal affective cues concealed in ordinary facial behavior [1]. Automatic ME analysis usually consists of spotting, which locates an event interval in a long video, and recognition, which assigns an emotion label to the detected interval [2]. This work focuses on spotting in long, untrimmed videos, where subtle ME motion must be distinguished from blinking, head movement, and acquisition noise.

Recent long-video ME methods have moved from window-based analysis toward video-level temporal modeling. Spot-then-Recognize establishes a continuous-video analysis framework, while ME-TST models temporal state transitions [1, 2]. ME-TST+ further models facial-region relationships, and BoostingVRME uses temporal priors for video-level regression [3, 4]. These methods produce a temporal response from an input video, but the response itself is not an event list. A downstream decoder must decide which candidate peaks are retained and how they are converted into event intervals. Although temporal localization explicitly models proposals and boundaries [15, 16, 17], long-video ME spotting must decode brief events from a frozen task-specific response.

Peak height alone is often insufficient for this decision. A high response may reflect a raised background, whereas a moderate peak can be more distinctive than its neighbors. Local structure provides complementary information, but smoothing can shift, merge, or suppress nearby peaks. Measurements at the same frame across smoothed responses therefore do not necessarily correspond to the same candidate. Candidate selection should account for both the response landscape and local temporal structure while preserving correspondence across scales.

We study candidate selection as a separate inference stage on frozen temporal responses. Global-Local Saliency Decoding (GLSD) forms reference-scale candidates and evaluates them with two signals. Normalized global prominence measures how strongly a peak stands out in the full response landscape. Peak-aligned local multi-scale saliency measures whether its bilateral local structure is supported after smoothing at nearby scales. GLSD filters candidates using these scores, while each upstream pipeline retains its own temporal scale, interval-construction rule, and conflict-handling procedure. This design allows the same decoder to operate on ME-TST+ and BoostingVRME responses without retraining their response generators. The main contributions are as follows:

- We establish candidate selection on frozen temporal responses as a distinct decoding paradigm, cleanly decoupled from response generation and interval construction.
- We propose GLSD, an elegant, training-free decoder synergizing normalized global prominence with peak-aligned local multi-scale saliency.
- Evaluated on SAMMLV and CAS(ME)³, GLSD boosts ME-TST+ and BoostingVRME F1 scores by 5.58%–18.15% while preserving their native event-output rules.

2. RELATED WORK

Long-video ME spotting has developed from handcrafted motion analysis to learned video-level representations. Early methods combine spatio-temporal fusion, optical-flow descriptors, or two-stream convolutional networks [23, 24, 25]. Earlier work also explored face-video heart-rate cues, recurrent optical-flow modeling, and multi-scale convolutional networks [29, 30, 31]. Subsequent work investigates scene flow and weak supervision to improve ME representations [5, 6]. Optical-flow features and multi-scale local transformers provide alternative ways to model short facial motion [7, 8]. M-pattern and duration-aware representations further describe the temporal evolution of expression events [10, 11]. These studies primar-

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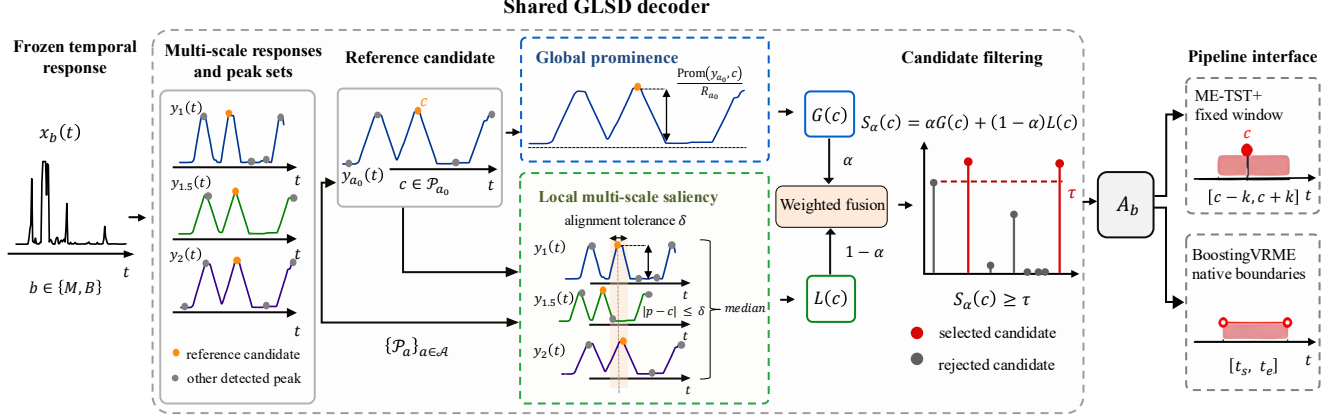


Fig. 1. The overall pipeline of the method. GLSD decodes a frozen temporal response by combining normalized global prominence and aligned local multi-scale saliency, then returns retained candidates through each pipeline’s native event-output rules.

ily improve the input representation or the temporal response generated by an upstream model.

The final event list also depends on the decoding procedure applied to a temporal response. Post-processing can change which candidate detections remain after suppression or conflict resolution [13, 14]. Scale-space analysis and peak detection provide tools for examining response structure over multiple temporal widths [22, 21]. GLSD builds on these observations at the response level: it scores reference-scale candidates by global prominence and aligned local evidence, then returns retained candidates to each pipeline’s native event-output procedure.

3. GLOBAL-LOCAL SALIENCY DECODING

3.1. Decoder overview and notation

GLSD is a training-free decoder for frozen temporal responses in long-video micro-expression spotting. Given a temporal response from an upstream pipeline, GLSD constructs reference-scale candidates, evaluates their global and aligned local structures, and retains candidates with sufficiently high scores. The retained candidates are then passed to the pipeline’s original event-generation interface. Figure 1 summarizes the shared GLSD scoring core and the pipeline-specific event decoding interfaces.

For pipeline b , let $\mathbf{x}_b \in \mathbb{R}^T$ denote the frozen temporal response of length T , where the temporal index t ranges from 0 to $T-1$. Let k be the base temporal scale in frames, and let A_b denote the native event-output interface. Let $a_0 \in \mathcal{A}$ be the reference-scale factor, $\rho > 0$ the local-radius factor, and τ the candidate threshold. We denote these decoding parameters by $\theta = (a_0, \rho, \tau)$.

3.2. Reference-scale candidates and global saliency

We examine the response at smoothing-scale factors $\mathcal{A} = \{1, 1.5, 2\}$. For each $a \in \mathcal{A}$, we round ak to the nearest integer and clip the result to $[1, T]$ to obtain the smoothing width w_a . The corresponding same-length, zero-padded moving-average response is

$$y_a = \text{ConvSame} \left(\mathbf{x}_b, \frac{\mathbf{1}_{w_a}}{w_a} \right), \quad (1)$$

where $\mathbf{1}_{w_a}$ is an all-one vector of length w_a , and ConvSame denotes same-length convolution. Duplicate rounded widths are retained once. At each retained scale, local peaks $\mathcal{P}_a$ are detected with minimum peak distance k and no height threshold. The reference-scale peak set $\mathcal{C} = \mathcal{P}_{a_0}$ defines candidate identities; peaks from other scales provide only local evidence for candidates in $\mathcal{C}$.

To compare scores across smoothed responses, the response range at scale a is

$$R_a = \max \left\{ \max_t y_a(t) - \min_t y_a(t), 10^{-12} \right\}. \quad (2)$$

For each reference candidate $c \in \mathcal{C}$, global saliency is the standard peak prominence on the reference response:

$$G(c) = \frac{\text{Prom}(y_{a_0}, c)}{R_{a_0}}. \quad (3)$$

Here, $\text{Prom}(y_{a_0}, c)$ denotes the standard prominence of peak c on y_{a_0} , evaluated over the full response without a restricted local window. Dividing by R_{a_0} makes global scores comparable to local scores across smoothing scales.

3.3. Peak-aligned multi-scale local saliency

Smoothing can shift a peak in time. GLSD therefore anchors each candidate at the reference scale and searches for a corresponding peak at every other scale before aggregating local evidence. The local radius is $r = \max\{1, \text{round}(\rho k)\}$, and the alignment tolerance is $\delta = \max\{1, \text{round}(0.5k)\}$.

For a peak $p \in \mathcal{P}_a$ at scale a , let $m_a^-(p)$ and $m_a^+(p)$ denote the minima of y_a over the clipped intervals $[\max(0, p-r), p]$ and $[p, \min(T-1, p+r)]$, respectively. Its normalized bilateral local saliency is

$$q_a(p) = \frac{\max \{0, y_a(p) - \max\{m_a^-(p), m_a^+(p)\}\}}{R_a}. \quad (4)$$

For candidate c and scale a , let $\mathcal{N}_a(c) = \{p \in \mathcal{P}_a : |p-c| \leq \delta\}$ denote the aligned peak set. If $|\mathcal{N}_a(c)| > 0$, GLSD selects its nearest

Algorithm 1 GLSD decoding of a frozen temporal response.

Require: Frozen response $\mathbf{x}_b$, base scale k , configuration $\theta = (a_0, \rho, \tau)$, fusion weight $\alpha \in [0, 1]$, scale set $\mathcal{A}$, and pipeline interface A_b

Ensure: Event set $\hat{\mathcal{E}}_b$

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1:  $r \leftarrow \max\{1, \text{round}(\rho k)\}$  and  $\delta \leftarrow \max\{1, \text{round}(0.5k)\}$ 
2: for all  $a \in \mathcal{A}$  do
3:   Compute  $w_a$  and  $y_a$  using Eq. (1)
4:   Detect  $\mathcal{P}_a$  and compute  $R_a$  using Eq. (2)
5: end for
6:  $\mathcal{C} \leftarrow \mathcal{P}_{a_0}$  and  $\mathcal{C}_\tau \leftarrow \emptyset$ 
7: for all  $c \in \mathcal{C}$  do
8:    $G(c) \leftarrow \text{Prom}(y_{a_0}, c)/R_{a_0}$ 
9:   for all  $a \in \mathcal{A}$  do
10:    Compute  $\ell_a(c)$  using Eq. (5)
11:   end for
12:    $L(c) \leftarrow \text{median}_{a \in \mathcal{A}} \ell_a(c)$ 
13:    $S_\alpha(c) \leftarrow \alpha G(c) + (1 - \alpha)L(c)$ 
14:   if  $S_\alpha(c) \geq \tau$  then
15:      $\mathcal{C}_\tau \leftarrow \mathcal{C}_\tau \cup \{c\}$ 
16:   end if
17: end for
18:  $\hat{\mathcal{E}}_b \leftarrow A_b(\mathcal{C}_\tau; k)$ 
19: return  $\hat{\mathcal{E}}_b$ 

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peak as $p_a^*(c)$; ties favor larger $q_a(p)$ and then earlier temporal order. At the reference scale, $p_{a_0}^*(c) = c$. The local evidence is

$$\ell_a(c) = \begin{cases} q_a(p_a^*(c)), & |\mathcal{N}_a(c)| > 0, \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

The aggregate local saliency is $L(c) = \text{median}_{a \in \mathcal{A}} \ell_a(c)$. Here, $\ell_a(c)$ is the local evidence contributed by scale a , and $L(c)$ is its median across all scales. The reference scale participates in the median. Thus, local evidence is associated with the same reference candidate across scales, while a missing cross-scale match contributes zero.

3.4. Global-local fusion and pipeline integration

For each candidate $c \in \mathcal{C}$, GLSD first combines the global and local scores as

$$S_\alpha(c) = \alpha G(c) + (1 - \alpha)L(c) \quad (6)$$

where $\alpha \in [0, 1]$ controls their relative contribution. GLSD then retains candidates whose fused scores meet the threshold:

$$\mathcal{C}_\tau = \{c \in \mathcal{C} \mid S_\alpha(c) \geq \tau\}. \quad (7)$$

To provide a common operating point across the two datasets, unless otherwise stated, $\alpha = 0.5$ is used in the main experiments.

For ME-TST+, each retained peak c maps to the fixed window $[c - k, c + k]$ clipped to $[0, T - 1]$; intervals are emitted in peak order. BoostingVRME preserves its native boundary search, candidate order, and conflict handling. Thus, candidate construction, structural scoring, and thresholding are shared, whereas final event geometry and output rules remain pipeline specific.

4. EXPERIMENTS

4.1. Experimental Setting

Datasets and evaluation. We evaluate ME-TST+ and BoostingVRME on SAMMLV [18] and CAS(ME)³ [19]. SAMMLV

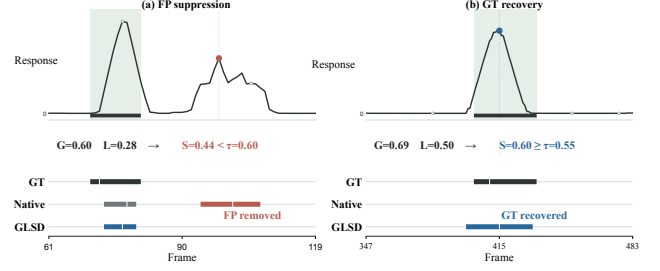


Fig. 2. Qualitative decoding on two real official-response cases. (a) Local cross-scale evidence suppresses a Native false positive. (b) Joint global-local evidence retains a Native-missed ground-truth event. Candidate scores come from the locked decoders; shaded regions mark ground-truth intervals.

Table 1. Native versus GLSD on official temporal responses. Relative gain is computed as $(F1_{\text{GLSD}} - F1_{\text{Native}})/F1_{\text{Native}} \times 100\%$.

Dataset	Method	Native	GLSD	Relative gain (%)
SAMMLV	ME-TST+	0.2723	0.2875	+5.58%
	BoostingVRME	0.2906	0.3088	+6.26%
CAS(ME) ³	ME-TST+	0.0912	0.0978	+7.12%
	BoostingVRME	0.0975	0.1152	+18.15%

contains 29 subjects, 79 videos, and 159 micro-expression events; CAS(ME)³ contains 94 subjects, 462 videos, and 853 events. Within each pipeline–dataset setting, Native and GLSD use the same official temporal responses, temporal units, interval-construction rules, conflict handling, and evaluation code. Spotting F1 is computed from TP, FP, and FN pooled across outer test folds.

Configuration selection. We use nested leave-one-subject-out (LOSO) validation. For each outer test subject, a configuration is selected by inner LOSO on the remaining subjects. The search space comprises 90 configurations, formed by three reference scales, three local radii, and ten candidate thresholds. For GLSD, we first retain configurations whose inner-LOSO F1 lies within $\epsilon = 0.002$ of the best value. Among them, we select the configuration that maximizes the smaller of its F1 gains over the global-only and local-only variants. Ties are resolved by GLSD F1, precision, fewer false positives, and a fixed configuration order. Paired subject-level 95% bootstrap confidence intervals are computed using 10,000 resamples.

4.2. Main Results

Quantitative comparison. Table 1 summarizes Native decoding and GLSD on identical official temporal responses. GLSD obtains higher pooled Spotting F1 point estimates in all four pipeline–dataset settings. The same pattern is observed for ME-TST+, which uses fixed-window output, and BoostingVRME, which uses native boundary search. Thus, the observed improvements are not restricted to one event-output rule. The paired bootstrap intervals in Table 1 quantify the subject-level uncertainty of these estimates.

Candidate-level examples. Figure 2 illustrates two official-response cases. A Native false-positive peak can have high amplitude but limited aligned local cross-scale support. Conversely, a ground-truth event can receive both global prominence and local multi-scale support. GLSD uses these signals to determine which response peaks are retained, while each pipeline preserves its native interval

Table 2. Comparison with published methods and our GLSD-enhanced variants (Spotting F1). S denotes spotting-only and S&R denotes joint spotting-and-recognition systems.

Type	Method	Year	CAS(ME) ³	SAMMLV
S	LGSNet [26]	2024	0.0990	–
	Li et al. [12]	2024	–	0.2541
	WCMN [9]	2024	–	0.2330
	Ofct [27]	2025	–	0.2466
	MSOF [28]	2025	–	0.2457
S&R	MEAN [1]	2023	0.0283	0.0949
	SFAMNet [5]	2024	0.0716	–
	ME-TST [2]	2025	0.0802	0.2167
	DiffME [20]	2026	0.0831	0.3036
	ME-TST+ [3]	2026	0.0912	0.2723
	+ GLSD (Ours)	Ours	0.0978	0.2875
	BoostingVRME [4]	2025	0.0997	0.2906
	+ GLSD (Ours)	Ours	0.1152	0.3088

Table 3. Ablation of the fusion weight α on ME-TST+ responses.

Dataset	$\alpha = 0$	$\alpha = 0.3$	$\alpha = 0.5$	$\alpha = 0.7$	$\alpha = 1$
SAMMLV	0.2697	0.2821	0.2875	0.2754	0.2807
CAS(ME) ³	0.0953	0.0978	0.0978	0.0980	0.0973

construction rules.

Comparison with published methods. Table 2 places the GLSD-enhanced pipelines alongside representative published long-video spotters. Published baseline values follow the comparison reported by BoostingVRME [4], while DiffME uses its reported result [20]. The GLSD rows use the current official-response protocol; these values should therefore be interpreted with the protocol differences noted in the table caption.

4.3. Ablation study

Locked-parameter component ablation. Table 3 examines the contributions of global prominence G and local multi-scale saliency L , together with fusion-weight sensitivity, on ME-TST+ responses. For each outer fold, we fix (a_0, ρ, τ) to the configuration selected for the main experiment in Table 1 and vary only α , without additional tuning. The endpoints $\alpha = 0$ and $\alpha = 1$ correspond to removing G and L , respectively.

GLSD achieves F1 scores of 0.2875 on SAMMLV and 0.0978 on CAS(ME)³ at the default setting. Removing G reduces these scores to 0.2697 and 0.0953, while removing L reduces them to 0.2807 and 0.0973. These results support the contributions of both components under the selected decoding configurations.

Hyperparameter sensitivity. We evaluate the effect of the fusion weight α in Eq. (6). For the interior values $\alpha \in \{0.3, 0.5, 0.7\}$, we retain the fold-level configurations selected for the main experiment, using the same outer subject splits, official temporal responses, and pipeline-specific event-output rules.

4.4. Cross-pipeline evaluation

GLSD shares candidate-level structural scoring while leaving event representation to each pipeline. ME-TST+ and BoostingVRME construct candidates from their own frozen responses and use the same

Table 4. Cross-pipeline transfer results (pooled Spotting F1). $\Delta = \text{Transferred} - \text{Native}$.

Dataset	Transfer	Native	Tuned	Transferred	Δ vs. Native
SAMMLV	M→B	0.2906	0.3088	0.2981	+0.0075
	B→M	0.2723	0.2875	0.2635	-0.0087
CAS(ME) ³	M→B	0.0975	0.1152	0.1166	+0.0191 [†]
	B→M	0.0912	0.0978	0.1008	+0.0095

[†] Paired subject-level 95% bootstrap CI excludes zero; the other three intervals include zero.

global-local scoring core, but retained candidates are converted to events by their native temporal scales, boundary rules, ordering, and conflict handling.

Table 4 summarizes the cross-pipeline transfer experiment, which tests whether a decoding configuration selected on one pipeline remains effective on another. For each outer fold, the source-selected (a_0, ρ, τ) is applied directly to the target pipeline on the same dataset, while the target retains its own temporal scale and event-output rules. Three of the four transfers improve over the target Native decoder. ME-TST+ → BoostingVRME is positive on both datasets, including CAS(ME)³ where the paired-bootstrap interval lies above zero, whereas the reverse transfer decreases on SAMMLV. These results suggest that the candidate-level scoring principle can transfer across pipelines, although the selected configuration remains response dependent.

4.5. Decoding Stability and Computational Efficiency

Perturbing the threshold τ selected in each outer LOSO fold by ± 0.05 changes Spotting F1 by at most 0.011 in every setting, and all perturbed results remain above their corresponding Native decoding results. The gains therefore remain stable around the selected operating point rather than being confined to a single threshold value.

GLSD operates only on the one-dimensional temporal response after response generation, using multi-scale smoothing, peak detection, alignment, and candidate scoring; it requires no additional network forward pass or visual feature extraction. The additional decoding time is 1.16 – 4.82 ms per video, excluding backbone inference, cache I/O, and offline configuration selection.

5. CONCLUSION

In this paper, we presented Global-Local Saliency Decoding (GLSD), a training-free candidate-level decoder for long-video micro-expression spotting. GLSD combines normalized global prominence with peak-aligned local multi-scale saliency to select reliable response peaks while preserving each pipeline’s native event-output rules. On official ME-TST+ and BoostingVRME responses from SAMMLV and CAS(ME)³, it improves pooled Spotting F1 point estimates in all four pipeline–dataset settings without retraining the response generators. The ablation confirms complementary contributions of the two cues, and the cross-pipeline results indicate partial transferability of the decoder. The locked-parameter fusion shows that the two structural cues contribute jointly to the GLSD decoder. Our code is public at <https://kx111666.github.io>.

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